



CO4301 – Individual Research Design/ Project

Project Proposal

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Research Project Title	Latency-Aware Deep Learning Framework for Real-Time Medical Image Segmentation
Key Objectives (Mention appropriately)	1 To design a latency-aware deep learning framework for real-time medical image segmentation. 2 To evaluate the proposed framework on complex medical imaging data, with angiography used as the primary experimental case. 3 To analyze accuracy–latency trade-offs in lightweight segmentation architectures.
Keywords (Maximum 5 separated by ;)	Medical image segmentation; Real-time deep learning; Latency-aware AI; U-Net variants; Efficient neural networks
Description (Describe the research concisely. Explain why your research is important) Brief about the planned methodology with proper justification. *If necessary give no more than 7 references from reputed sources. *Do not exceed 3 pages	<u>Description</u> Medical image segmentation is a fundamental task in computer assisted diagnosis, enabling automated identification of anatomical structures and regions of interest. Despite recent advances in deep learning based segmentation, many state-of-the-art models remain computationally intensive and unsuitable for real-time clinical environments where low latency and stable performance are essential. This research proposes a latency-aware deep learning segmentation framework that explicitly integrates inference time constraints into model design and optimization. Instead of focusing solely on segmentation accuracy, the study prioritizes the balance between accuracy and computational efficiency, which is critical for real-time clinical use. The framework will be evaluated using angiography images, which present challenges such as low contrast, overlapping vessel structures, motion artifacts, and noise. Robustness will be examined by testing segmentation performance and inference time stability



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	under varying angiography image conditions without modifying the network architecture. By emphasizing deployability and efficiency, this study aims to provide practical insights into real-time angiography image segmentation for clinical workflows. <u>Brief about the Planned Methodology with Justification</u> Data Collection and Preprocessing: Publicly available medical imaging datasets will be used, with angiography serving as the primary dataset for model development and evaluation. Preprocessing steps will include spatial normalization, intensity normalization and contrast enhancement to address modality specific imaging challenges and improve segmentation stability. Model Development: A baseline U-Net architecture will be implemented, followed by the development of lightweight and efficiency optimized variants. Architectural modifications will include: • Depthwise separable convolutions • Reduced encoder–decoder depth • Channel width optimization These modifications are guided by the objective of minimizing inference latency while maintaining acceptable segmentation accuracy. Training and Evaluation: The models will be trained using supervised learning with loss functions such as Dice Loss and Binary Cross-Entropy Loss. Evaluation will focus on both segmentation quality and real-time performance, measured using Dice Coefficient, Intersection over Union (IoU), inference latency in milliseconds per image and frames per second (FPS) under fixed hardware conditions. All models will be evaluated using the same input resolution and testing protocol to ensure fair comparison. Performance results will be compared against a baseline U-Net architecture to clearly quantify accuracy–latency trade-offs introduced by efficiency oriented architectural modifications. Justification and Expected Contribution: The proposed research addresses a key limitation in existing medical image segmentation studies by explicitly treating inference latency as a primary design constraint rather than a secondary
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performance metric. By systematically analyzing accuracy–latency trade-offs in lightweight segmentation architectures, the study provides practical guidance for designing models suitable for real-time clinical environments. The findings directly support the development of deployable medical AI systems and offer experimental insights that can be reused in larger system-level implementations.

References

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